Cthulhu's cat sitter research

Firstly, I think it is important to evaluate the main themes and key components that will make up the game and list them down. This was thought as a basis for research into similar games relating to gameplay, theme, and platform.

- Non-Euclidean
- VR
- early 2000s/original PlayStation style of graphics
- single player
- puzzles
- 1990-2010 stereotyped American home
- Horror elements

Expansion of bullet points then suggested areas further development:

Euclidean geometry refers to a mathematical system originally postulated by Euclid and reflects what we see in the real world. The system relies on the following axioms:

(From Wikipedia)

1. To draw a straight line from any point to any point.
2. To produce (extend) a finite straight line continuously in a straight line.
3. To describe a circle with any centre and distance (radius).
4. That all right angles are equal to one another.
5. [The parallel postulate]: That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which the angles are less than two right angles.

What we can derive from this is the system of geometry that we use to describe the world around us, so naturally deviating from Euclidean geometry leads us to create an environment that does not reflect the real world right? Well not exactly. Euclidean geometry makes one major assumption and that is that we are dealing with a flat space. This may seem like a deviation from relevant research however is an important distinction to make. Simple example of where Euclidean geometry no longer applies is when you go to construct a square on the surface of a sphere. Connecting four lines of equal length with congruent angles will give you a square with 120° angles. It also becomes possible to construct a five-sided square with equal side lengths and 90° angles providing you construct it on the correctly curved surface. A reason I wanted to make this distinction is because Euclidean geometry can fall down at something as simple as navigating about one of the poles, say you start at the North Pole walk 100 feet South, 100 feet east then 100 feet north and you'll end up in the same place that you started. Euclidean geometry with demand that you ended up 100 feet east of where you started.

When we think of non-Euclidean geometry, we may immediately jump to games such as portal and portal 2. Though somewhat famously non-Euclidian, if we also ignore the portals,

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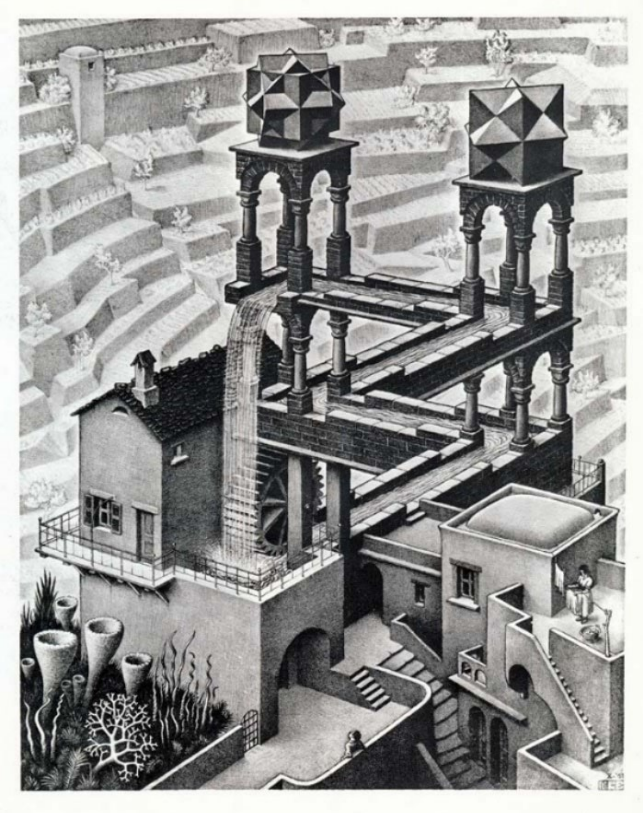
portals game world follows Euclidean geometry perfectly. This allows the portal gameplay itself to be a large part of the players focus, this is largely due to the fact that the rest of the environment is familiar to the player. Portal also likes to introduce new mechanics and elements of gameplay to the player slowly and is renowned for striking a good balance between not overwhelming the player but also maintaining engagement through the use of more challenging puzzles and combinations of different mechanics. The non-Euclidean elements within portal, i.e., the portal gun, are used primarily as tools to solve puzzles rather than as part of the environment, or as the setting of the puzzles themselves.



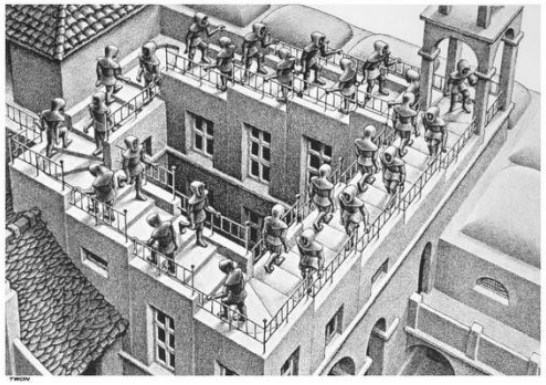


In order to create more immersive environment for game that is less story driven than portal, some other games make the environment itself Euclidean. This can be done in many different ways so before looking at specific games I will list some examples:

- That Tardis from Doctor Who
- Penrose stairs
- A waterfall that feeds itself (M.C. Escher)
- A portal that allows you to teleport through space
- Objects that change shape based on position







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Superluminal makes use of forced perspective and optical illusions to create a surreal puzzle game. One of the main elements of the game is the ability of the player to pick an object up and hold it in front of them, for example dice, by holding the die in front of you such that is looks table sized across the room and releasing it, the die would become that size.





Giant chess pawns feature throughout the game and sometimes puzzle elements can be hidden withing the furnishings of the level such as using a chair or an obeject that apears out of thin air when you lign up a perspective painting of it. This style of mechanic forces players to explore levels and interact with their surroundings to help solve the puzzles, acommon trope between many puzzle games including the test chamber sections of portal.

I believe the reason that very few fully/largely non-Euclidean games exist (espetially any with a high production value) is due to how disorienting and confucing it can be, additionally it would be hard not to fall into the trap of being gimmicky. There are several very intersting games out there but they feel like a technical demonstration more than a game. A combination of technical challenges and difficulty with regards to having a compelling narative of any kind limits the appeal of fully non-Euclidean games to developers. Largey I believe this is a good thing due to the difficulty of working with an existing engine for such a project.

For these reasons I beilve our best approach is to limit the non-euclidean elements to trickery such as teleporting between almost identtical sceens for example, or a mirror that shows objects slightly diffently to how they appear unreflected.

VR is still a relitivly forein medium to many gamers despite its growth, for instance I had never played a vr game before starting this research. I think it is important to experiece vr first hand in order to write about it with any acuracy, I chose Half-Life Alyx as it received rave reveiws when it launched and I have a vague idea of the game world from watching and experienceing gameplay from other games from the same universe. Things I would like to note are:

- What I expected before I started
- Things that surprised me initially
- What VR does well
- Pitfalls of current VR tech
- Mechanics and design of the game

I was expecting VR to feel fairly disconcerting, after all without being able to see or hear your surroundings well due to the headset and headphones I expected it to be difficult to walk around without worrying about hitting something in the relatively small space I had

access to. Another aspect I was unsure of was how hard it would be to learn to use the controllers when you can't see them. The HTC VIVE has fairly simple controllers with a rear trigger, side paddle, top buttons and a large touch sensitive D pad. The layout appears fairly intuitive, but I expect finding the top buttons to be a bit of hassle without being used to the controllers.

What surprised me initially was only using two fingers for the controls. I believe this is due to the fact that I transitioned to playing games on pc around five years ago and am currently most comfortable with mouse and keyboard. As an experiment I picked up my PlayStation DualShock controller to see how it sits in my hands. I noticed that I actually end up with the same feeling, in fact it's worse, personally I prefer to use my index finger for both shoulder buttons and the triggers meaning that I have three fingers sat underneath simply gripping the controller, I can certainly see why there are multiple options for third party controllers that feature multiple rear paddles. I would be very interested to compare the feel of valve's index controllers as they strap to the hands and track the fingers without buttons meaning that they provide more options for interacting in a game but in a less tactile manner.

Another thing that surprised me was the way VR deals with how much space you have available, by tracing out your environment, virtual walls are constructed that appear as a fence when you get close to them, as you back away they fade and stop being visible, this allows you to be confident that you aren't going to walk into a wall for example. Another similar thing is that the controllers are motion tracked so its easy for them to be rendered in front of you helping make it easier to learn the controls of a game.



What VR does incredibly well is immersion. There are a lot of pretty non-VR games out there, even slightly older games like Battlefield V and Shadow of the tomb raider (both nearing half a decade old) hold up well on most people's hardware and some more recent

games are truly breath taking but VR does something that you physically can't replicate with a flat screen. Immersion. When you first load into Half Life Alyx, you find yourself on a roof terrace in an eastern European city, on pc you would be able to look around and walk about but you'd probably leave the area in about 30 seconds flat. In VR there is a lot more to explore, you can walk up to the railing and lean over edge and look into the street below, I found myself enthralled by, of all things, the portable radio they place on the table next to you, being able to physically interact with the radio and see the fans turning as 3d objects is a fairly major departure from games on pc. I liken this to the difference between analogue and digital, take a puzzle that requires you to rotate wheels for example. In a pc game you would hit an interact key and the wheel would rotate so that a specific symbol lines up, in VR however you can spin the wheel at speed, slowly rotate it by hand, give it a gentle push, whatever you feel like doing. Whilst I appreciate it is possible to do this kind of thing in a pc game using a combination of keystrokes, it is in no way as organic and natural feeling.

The main downfall of VR currently is being able to justify it as a consumer. Quite frankly headsets aren't there yet. There are some very promising offerings that promise high refresh rates, high resolution, wides field of view and a lightweight package but currently nothing ticks all the boxes and any that come close are prohibitively expensive. What makes VR even more expensive is the requirement to have a powerful enough computer to run it. You are looking at an extra £600-1200 investment to play VR games, sure there are cheaper options but the only decent offering here is by meta and has some very dubious privacy concerns as an additional cost instead. For most gamers is a simple equation, buy a better GPU and save money on the difference. Most pc gamers would rather spend the money on performance or peripherals that they use all the time. Another downfall of VR is that it has a niche use case, a nice mechanical keyboard is great for gaming but it's also nice to type on and can be used for work as well, VR is yet to see widespread use for productivity which means it's purely used for gaming and it makes the justification to buy a headset even harder, especially since the technology is still maturing. Focusing back on the actual headset for a moment, I found it confusing trying to adjust eye spacing, it would be nice if this was explained during setup. Automatic IPD is another currently high-end feature that we will have to wait to trickle down into affordable headsets.

Let's actually talk about Half Life now, going into it, I didn't really know what to expect, I don't know a huge amount about the lore surrounding the game but the world itself is fairly familiar to me. After a little bit of navigating the city to get used to the controls and feel of the game you are introduced to a bit of story, and you meet Russell face to face where you acquire two very important things. A gun, and the gravity gloves. The gravity gloves allow you to manipulate gravity for a given object, this allows you to gesture toward an object, flick your wrist (the object is launched toward you) and catch the object in your hand. The gun you are provided with is a fairly standard looking pistol somewhat similar to a Colt M1911 (a few notable quirks that I'll skip over) with a few sci-fi elements and a weird trigger guard to make it look up to date. But something magical happens when you combine these two, you get the most satisfying gameplay loop I have ever experienced in any game. The loop starts when you need to reload your gun. (If you're shooting right-handed) first, using

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your left gravity glove, launch a magazine from the environment around you towards you and catch it, now immediately discard the empty mag from your pistol, insert the new mag with your left hand, rack the slide then start shooting. What is happening is you are witnessing a pretty cool reload animation like most shoots except because its VR, its not an animation, you are in control the whole time. This is an example of what is probably the best thing about what VR controls allow.







Late 90's/early 2000's graphics are surprisingly varied. 2d games typically look more polished due to the limited processing power required to render them compared with 3d games but it would be wrong to neglect how genuinely impressive some of the 3d graphics of the time were. Growing up I played the 2010 remake of goldeneye on Nintendo Wii quite a lot, what is honestly quite surprising to me is comparing the graphics of the original next to the remaster and then the remake. The remaster is simply a graphics bump which is to be expected but what I found most interesting was the jump to the remake where the extra performance wasn't focused so much on graphics but on physics. My key take away from this is that sure, early 2000's games are low poly, but not as low poly as you would expect. Keping the environment low poly allows for more processing for better character models and props.

I think Grand Turismo 2 is a good reference for exterior and environmental design and a selection of different games should be used for the interior, I say this because it's difficult to find much refence material that would cover all of our required interior elements.

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Stereotypical American home. The first thing I immediately jump to is the COD map Nuketown. As the name suggests the map comprises of several homes and a few quintessentially American vehicles. As the name suggests it is designed to mimic one of the mock towns that the United States built in the desert when they were testing the effects of their nuclear weapons and what they could expect to happen should an enemy nuke ever be set off in the US. I believe the main reason that I see this as a typical American house is largely due to the boxy, wood framed construction of a two Storey house with a shingled roof and a double garage tacked on the side.



Due to how varied the United States is there are quite a lot of different options when it comes to picking a recognisable style of house. Different influences including colonial architecture that consists of low single story buildings with relatively steep sloped roofs, often featuring wood cladding; ranch style houses that typically feature along open layout contained on a single story; Houses that take Italian inspiration; Victorian style houses typically relatively narrow yet tool with Bay windows and decorative woodwork; there is also the style the Americans referred to as a cottage which naturally differs greatly from what you would think of when referring to a British cottage.



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The style we are looking for is that of the Nevada test site (the mock town that Nuketown is based off of). The fake town is based off of the 50's/60's idea of the quote 'American dream'. What this means in real terms is a big house with a green lawn and a driveway with a nice car on it. The shear scale of sububan sprawl in the US immense, the size of these areas in America compared to the structure of European cites is staggering, the key difference is that in America they are all detached houses rather than the compact brick terraces and tower blocks you see all over England.





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Horror elements:

- Isolation
- Lighting
- Music and sound effects
- Unexpected elements
- Feeling of being chased
- Jump scare elements
- Being powerless
- Tension

Isolation should be relatively easy to achieve due to the fact that this is a single player game, the environment also plays an important role in this, being in a location where you would expect to find people and there being no signs of life would help with this.

Dim, warm lighting in combination with the colour palette and large shadows help create an environment where you have a limited ability to see parts of the space around you and you subconsciously question if there is something there that you can't see.

An otherwise definitely quiet environment with loud sound effects makes them stand out and also emphasises the silence and isolation. Relatively simple music consisting full example of solo piano provides you with limited accompaniment and lacks any reassurance.

An unexpected element could be something as simple as a dead end where you wouldn't expect one boy suspiciously out of place item that leads you to question whether or not it's apparent is something that should concern you. You could also have an object that appears or disappears the second time that you enter a space as this would make you question whether or not you were seeing things.

The feeling of being chased can be achieved through the use of lighting for example a shadow that appears to gain on you as you move. It could also be achieved through the use of sound, footsteps that gradually get closer for example objects being knocked off of shelves behind you, the sound of something rolling along the floor slowly etc.

A jump scare element that I think would work quite well in VR would be something such as turning a corner that you've already navigated and being confronted by a mirror for life sized portraits, hiding members within the environment would provide you with a moving object in your peripheral vision should help act the sense not being alone.

A Feeling of powerlessness can be achieved quite well by removing the players ability to control the environment, giving them a slowness effect or, in a non-Euclidean world, having the corridor keep extending as you try and run through.

A good way to build tension would be to add time pressure that slowly builds, think of the theme used on the show 'countdown'.